

Sia Zhong Tai

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Available to start from July 2026

EDUCATION

UNIVERSITY TUNKU ABDUL RAHMAN (UTAR)

Bachelor of Software Engineering (Honours) | 2023 – *Present*

CGPA: 3.69

Expected Graduation: September 2026

Awards: President's List (May 2023), Dean's List (June 2024, February 2025, June 2025)

WORKING EXPERIENCE

Software Engineering Intern | *GenTech AI Sdn Bhd* (Oct 2025 – Jan 2026)

- **AI Agent Architecture:** Designed and deployed a stateful Multi-Agent System using LangGraph and FastAPI, architected to scale across 15–20 Trello boards for a 10-member team — implementing an "Ask-and-Listen" workflow that proactively surfaces project blockers and automates daily progress collection without manual status updates.
- **Low-Code Form Engine:** Architected a JSON-Driven Form Engine (React/TypeScript) as the single source of truth for 8 Malaysian Tax Form C sub-forms, implementing a recursive renderer with declarative validation and Excel-like formula evaluation. Enabled non-technical staff to modify complex tax logic and export internal compliance reports as PDFs without code deployments.
- **Resilience Engineering:** Independently recovered 2 weeks of lost engineering work caused by a faulty merge sequence — ruled out cherry-picking as infeasible, then used LLM-assisted Git diff analysis to reconstruct every lost change and reimplement them in full within 2 days, with zero feature regression.

PROJECTS

AI Academic Assistant (Final Year Project) | *Next.js, React Native, FastAPI, Supabase, LangGraph, Python*

- Shipped a full-stack AI academic platform across 4 repositories — cross-platform frontends, a production Supabase backend, and a Python AI microservice housing 6 fully implemented LangGraph agents.
- **Adaptive Learning Engine:** Consumes semantically chunked content to drive quiz and summary generation via a 3-Phase Chain-of-Thought pipeline with auto-correction loops. Mastery is tracked per-concept using EMA scoring, feeding into FSRS spaced repetition scheduling to surface the right review material at the right time.
- **Autonomous Scheduler:** Built a ReAct orchestrator agent invoking Google OR-Tools CP-SAT solver to dynamically fit tasks into calendar slots under hard constraints (fixed class times) and soft constraints (user-preferred study hours), with automatic missed-session rescheduling triggered by the daily background cron.
- **LMS Ingestion Pipeline:** Engineered an async ingestion agent with SHA-256 content

deduplication, LLM-powered semantic chunking with difficulty-tier assignment, and DAG knowledge graph extraction via networkx — serialised as JSONB and rendered on the frontend with a dagre layout for topic visualisation.

- **Background Infrastructure:** Eliminated external scheduler dependencies by designing a 3-loop asyncio cron system (60s dispatcher, hourly sync, daily 06:00 AM per-user planner) running natively within FastAPI lifespan — managing FCM push notifications, FSRS quiz planning, and overdue session rescheduling without Celery or Cloud Scheduler.
- **Security Architecture:** Enforced Row Level Security across all database tables, Supabase Edge Functions as a Bearer-authenticated AI gateway, and signed URL file storage — ensuring zero cross-user data exposure at the database layer.

AI-Powered Outdoor Event System | React 19, Firebase, Tailwind CSS, Vite PWA

- **Real-Time Event Architecture:** Deployed a server-authoritative PWA used live by 6 team managers across a 60-participant outdoor event. Engineered Firestore real-time subscriptions with timestamp synchronization to prevent clock drift across concurrent devices.
- **Concurrency & Game Logic:** Implemented atomic territory capture logic using Firestore Transactions (runTransaction) to eliminate race conditions under concurrent writes, integrated with a weighted scoring and tiered progression algorithm updated in real-time.
- **Role-Based Access Control:** Designed a three-tier RBAC system (Manager, HQ, Admin) using Firebase Anonymous Auth, assigning role-scoped tokens at authentication to enforce distinct trust boundaries and restrict workflow permissions at the application layer.

TECHNICAL SKILLS

- **AI & ML:** LangGraph, LLM Integration, RAG Pipelines, Prompt Engineering, n8n
- **Backend:** Python, FastAPI, Supabase (PostgreSQL, Edge Functions, Realtime, Storage), Node.js, MongoDB, REST APIs, Microservices
- **Frontend:** TypeScript, Next.js (App Router), React Native (Expo), Vue.js, Tailwind CSS, shadcn/ui, Zustand
- **DevOps & Tools:** Docker, GitHub Actions, CI/CD, Git, Firebase (Firestore), LangSmith, Vercel
- **Languages:** Python, TypeScript, JavaScript, SQL, C++, Java

CERTIFICATIONS

Microsoft Azure AI Fundamentals (AI-900) — 2024

Purdue University — Machine Learning in Action (Online Certification) — 2024

LEADERSHIP & ACTIVITIES

Galaxy Chamber Orchestra — Concertmaster (2023 – Present)

Led operations and logistics for concert events with 100+ performers. Long-term committee member since 2019; active member since 2012.

Program Akar Powered by Microsoft — Cohort 2 (2025)

Awarded 2nd Place in PayNet's national fintech ideathon, competing in an AI-driven financial solutions challenge.